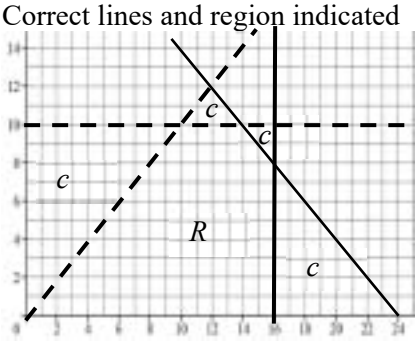
Question	Answer	Marks	Partial Marks
8(b)	Correct lines and region indicated 	**6**	**B1** for each correct line and **B2** for R in correct region for all 4 correct lines or **B1** for R in any one of the regions marked *c* or **B1** for R that satisfies 3 of the correct inequalities
8(c)	228 nfww	**2**	**M1** for $8x + 12y$ for any (x, y) in *their* R, x, y both integer or $x = 15$, $y = 9$
9(a)	[a =] 9 [b =] 14	**3**	**B2** for a =9 OR **M2** for $\frac{60}{360}\times 2\times\pi\times 17+\frac{60}{360}\times 2\times\pi\times 10+7+7$ oe or **M1** for $\frac{60}{360}\times 2\times\pi\times 17$ oe **or** $\frac{60}{360}\times 2\times\pi\times 10$ oe If 0 scored **SC1** for b =14
9(b)(i)	60° at centre or interior angle = 120°	**B1**	
	[6×]$\frac{1}{2}\times d^2\times\sin 60$ oe	**M1**	
	[d^2 =]$\frac{127.3}{6\times\frac{1}{2}\times\sin 60}$	**M1**	
	6.99[9…] to 7.00[…]	**A1**	Dep on M1M1
9(b)(ii)(a)	1273	**1**	
9(b)(ii)(b)	675 or 674.5 to 674.6	**2**	**M1** for 2 ×127.3 oe or 6 × 7 × 10 oe
10(a)(i)	(4.5, −1)	**2**	**B1** for each

Question	Answer	Marks	Partial Marks
10(a)(ii)	6.71 or 6.708...	**3**	**M2** for $(6 - 3)^2 + (2 - -4)^2$ oe or better or **M1** for $[-](6-3)$ and $[-](2--4)$ oe or for $([-]3)^2$ and $([-]6)^2$ oe
10(b)(i)	$-\frac{4}{3}$	**2**	**M1** for $3y = -4x + 12$ or $\frac{4}{3}x + y - \frac{12}{3}$ [= 0] or better seen
10(b)(ii)	(0, 4)	**2**	**B1** for each or for $y = 4$ not in coordinate form
10(b)(iii)	$[y =]\frac{3}{4}x + \frac{1}{2}$ final answer	**3**	**M1** for gradient $\frac{3}{4}$ or $\frac{-1}{\textit{their}\textbf{(b)(i)}}$ oe or better **M1** for (6, 5) substituted into $y = \frac{3}{4}x + c$ or $y = \textit{their } mx + c$ oe
11(a)	$-4(-1)^3 - 9(-1)^2 + 5$ or better	**M1**	
	= 0 [so stationary point]	**A1**	with no errors
11(b)(i)	18	**3**	**B2** for $6x^2 - 6$ isw OR **B1** for $6x^2 + k$ (any k) isw or $px^2 - 6$ isw ($p \neq 0$) or $6x^2 - 6 + 8$ **M1**dep on B1 for $x = -2$ substituted into *their* $\frac{dy}{dx}$
11(b)(ii)	1 and –1	**2**	**M1** for $6x^2 - 6 = 0$ oe seen or for *their* $\frac{dy}{dx} = 0$ if B1 scored in part **(b)(i)**

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